

Causal Localization and Shared Region Search for Bounded Pachner Simplification

A fixed-parameter step toward quasi-polynomial unknot recognition

Research continuation for Vladimir Reshetnikov's ProveIt project
Proofs, algorithms, experiments, and integration artifacts

9 October 2026

Abstract

The current ProveIt work already searches all small connected regions of a triangulation and compresses commuting Pachner traces. Its remaining locality hypothesis can be removed for the specific objective of decreasing the number of tetrahedra. We prove that if a generalized face-pairing triangulation has any simplifying 2–3/3–2 sequence using at most U total upward moves, then it has a first-descent sequence of length at most $2U + 1$ whose actual original footprint is connected and contains at most $3U + 3$ tetrahedra. The proof factors traces through a graph of tetrahedron births and consumptions; a minimum-length witness makes its additive loss concentrate in one component. A six-tetrahedron ball proves sharpness at $U = 1$.

Together with an explicit constant-alphabet trace encoding, this yields a deterministic $2^{O(U)} \text{poly}(t + B)$ algorithm for bounded-total-upward simplification, where t is the source size and B bounds the binary marking. Thus logarithmic U gives polynomial time, and $U = O(\log^2 t)$ gives $t^{O(\log t)} \text{poly}(B)$ time. A further localization and deterministic color-coding argument decides reduction by at least k tetrahedra within the same joint upward budget in $2^{O(U+k)} \text{poly}(t + B)$ time. We state the additional structural requirements for complete recognition.

The implementation replaces independent restarts on overlapping allowed regions by one exact search with an inverted cover index, lazy move production, and independent source-bound replay. A companion extractor shortens supplied descent certificates while preserving all local frames admitted by the verifier. The accompanying experiments retain exact endpoint comparisons, genuine solid-torus controls, a sharp obstruction to a smaller search radius, raw paired timings, and adversarial certificate tests. These results establish bounded simplification and reusable search improvements; a general quasi-polynomial unknot recognizer still requires a global structural and terminal theorem.

Reproducibility baseline. 8188525b70033dcfe7c51ea5ae2c8723ad0c0198

Source tree. [ProveIt](#) / [Topology](#) / [UnknotRecognition](#)

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1 Result, motivation, and source audit

The principal advance is a locality theorem for *tetrahedron descent*. Previously, a small connected region was an additional hypothesis of the parameterized search guarantee. Here its existence follows from the upward budget and the objective itself. This is useful because the input to a simplifier normally specifies a triangulation and a resource budget, not the location of the successful modification.

1.1 The bounded question that is completely answered

Write $|T| = t$ for the number of tetrahedra in a generalized face-pairing triangulation. An upward move is a legal 2–3 replacement and a downward move is a legal 3–2 replacement. The decision problem is

$$\text{Does there exist a legal } \sigma : T \rightsquigarrow T' \text{ with } u(\sigma) \leq U \text{ and } |T'| < t? \quad (1)$$

The budget counts *all* upward moves, including upward moves made after earlier decreases. It is not a bound on the largest intermediate triangulation.

For this precise question, the article proves an unconditional fixed-parameter algorithm in the generalized move model. The implemented version accepts the repository’s finite, compact, connected, orientable triangulations with one torus boundary and an integral cocycle. The combinatorial locality proof itself applies to the corresponding formal relative-boundary moves on other valid generalized 3-manifold triangulations as well. Ideal inputs are not thereby accepted by the native finite-manifold validator.

The central conclusions are as follows.

1. A successful trace has a successful replacement using $u \leq U$ upward moves, $u + 1$ downward moves, and at most $3u + 3$ original tetrahedra. Its actual original footprint is connected in the original dual graph.
2. Complete search for (1) takes $2^{O(U)} \text{poly}(t + B)$ bit operations. The single-exponential dependence follows from explicit trace encoding, not from treating the set of geometrically possible triangulations as small.
3. Reduction by at least k tetrahedra under a joint upward budget is decidable in $2^{O(U+k)} \text{poly}(t + B)$ time, using bounded total footprint, disjoint-trace composition, and deterministic color coding.
4. The region-union search preserves the full declared bounded endpoint family. It shares common prefixes, indexes exact cover membership, and materializes only transitions actually visited.
5. A valid input descent certificate can be projected and truncated to a connected first descent, then independently replayed. This operation requires preserving the declared local vertex frames of born tetrahedra.

The cost bound includes polynomial-time source validation, move generation, transport, and the elementary endpoint test $|T'| < t$. An arbitrary disc-finding predicate has its own cost and does not inherit the component-projection argument merely because it is queried by the same search engine.

1.2 What was already present

The repository was read through its selected GitHub connection and cloned at the fixed revision

8188525b70033dcfe7c51ea5ae2c8723ad0c0198.

The three relevant archives in docs/incoming at that revision were:

- `unknot_trace_search_20261009.zip`;
- `unknot_active_faces_batched_descent_20261009.zip`;
- `unknot_minimum_envelopes_20261009.zip`.

Their article sources and code were examined before choosing this continuation. They are preserved as prior work, not reported as results first obtained here.

The trace-search archive already contains both an automatic canonical-parent enumerator for connected original regions and the bound

$$t^c 2^{O(r+U)} \text{poly}(t + U + B), \quad (2)$$

where the actual consumed original cells are contained in a permitted region of size at most r with at most c connected components. It also proves the generic 11^{3r+5U} commitment bound and implements exact sleep-set exploration [2]. Re-proposing that region enumeration would duplicate the incoming result. Our new implication is that successful one-unit descent automatically admits $c = 1$ and $r \leq 3U + 3$.

The active-face archive studies normal-sector rank, support propagation, and simultaneous selection of already supplied disjoint coherent collapses [3]. Those results address a different optimization problem: selecting or evaluating a batch whose candidates have already been produced. The minimum-envelope archive studies exact sector optimization [4]. Neither supplies the causal-localization statement proved below.

The working baseline retains all current native files. Only files absent from the native tree were imported from the trace archive. Six common files differ between the current native tree and that older archive; the native versions were retained. These include the current unit-ray disc kernel and topology observers. The integration manifest records byte hashes, origins, and the precise new overlay. No result depends on silently replacing current native code with a historical snapshot.

1.3 Relationship to the literature

The generalized face-pairing model and the local legality of 2–3 and 3–2 moves are standard in computational 3-manifold topology. Burton and He explicitly describe the distinct-tetrahedron condition and the internal degree-three edge criterion; they also discuss tracking objects through renumbering [5]. The present proof makes the birth/consumption relation sufficiently explicit to factor simplifying traces and bound their original support.

Dependency DAGs and topological reorderings are established tools in parameterized geometric search. Kanj and Xia use them in their fixed-parameter algorithm for planar flip distance [6]. That is a methodological precedent, not a theorem about these 3-dimensional replacements. The local legality and relative-boundary arguments needed here are proved separately. We claim an extension relative to the supplied repository; priority for an equivalent result in the entire literature is not established by the targeted source search.

Lackenby’s author-level 2021 handout states the quasi-polynomial recognition bound

$$2^{O((\log n)^3)} = n^{O((\log n)^2)}$$

and describes a hierarchy-based route [7]. This should be distinguished from the more specific $n^{O(\log n)}$ target in the project. The July 2026 preprint gives a new incompressibility and unknot-recognition algorithm, but its discussion of running time does not furnish the missing quasi-polynomial implementation bound. In particular, Proposition 9.1 bounds iterations by $L(g+1)^L$ in hierarchy parameters, and the paper explicitly discusses further acceleration [8]. A bounded Pachner simplifier does not reproduce that hierarchy algorithm.

2 Formal replacements and encoded states

2.1 Tetrahedra, ports, and formal boundary triangles

A generalized triangulation consists of finitely many abstract tetrahedra, each with local vertices $0, 1, 2, 3$, and affine pairings of some triangular faces. A pairing is stored by the partner tetrahedron and a permutation of the four local vertices. Unpaired faces form the boundary. The quotient is required to be a valid manifold of the declared kind. Different edges may have the same global endpoint vertices; face pairings within one tetrahedron and repeated adjacencies are permitted when manifold validation accepts them.

Both moves replace a formal bipyramid while preserving its six individually identified boundary triangles. Give its five formal vertices the labels $0, 1, 2, 3, 4$. The two-cell filling is

$$(0, 1, 2, 3), \quad (0, 1, 2, 4),$$

and the three-cell filling is

$$(3, 4, 0, 1), \quad (3, 4, 1, 2), \quad (3, 4, 2, 0).$$

The upward move changes the first filling to the second. Its two old tetrahedra must be distinct. The downward move changes the second to the first; its central edge must be interior, have exactly three local edge occurrences in three distinct tetrahedra, and have the compatible triangular link of the formal bipyramid.

The boundary attaching permutations are retained even when some boundary faces are attached to each other or their vertices are globally identified. This contract is stronger than merely saying that the old and new regions have homeomorphic boundaries. It preserves individual boundary ports and is what supports the causal proof.

Definition 2.1 (Incarnations and traces). A tetrahedron *incarnation* is a tetrahedron from its creation to its unique possible consumption. An original tetrahedron starts with its own source identity. A new incarnation is named by its producing event and its output slot, with its local vertex labels retained. A trace is an ordered sequence of legal replacements together with this tracking data.

If a trace contains u upward and d downward events, then

$$|T'| = t + u - d. \tag{3}$$

We write $\text{loss} = d - u$. A strict descent means $\text{loss} > 0$.

2.2 Original footprints and covers

Let G_T be the simple dual graph on the original tetrahedra: two vertices are adjacent when the tetrahedra share a paired face. Its maximum degree is four; loops and repeated face adjacencies do not increase that degree.

Definition 2.2 (Actual footprint). The actual original footprint $F(\sigma)$ of a trace σ is the set of original tetrahedron incarnations it consumes. An allowed region A is a *cover* when $F(\sigma) \subseteq A$. Original tetrahedra outside A are protected, while every born incarnation remains eligible.

An allowed connected cover may contain unused cells, and its consumed subset may be disconnected. The earlier region search uses covers. The new localization theorem is stronger: its selected witness has a connected *actual* footprint, so no auxiliary connecting corridor is needed.

2.3 Markings and the implementation boundary

The implemented source carries integral height rows $h_t = (h_{t,0}, \dots, h_{t,3})$ whose differences agree on identified oriented edges. Adding a constant to one row does not change its local edge values. Rows are normalized after transport. The move is selected by geometry; the cocycle is carried through the same formal replacement.

Throughout the complexity statements, B bounds the bit length of the supplied height entries and source edge values, up to a fixed additive constant. Bounding differences alone would omit the cost of reading arbitrarily large common offsets in unnormalized input rows. The source itself has $O(t \log(t+1))$ combinatorial bits. Explicit snapshot certificates include intermediate face pairings and height rows. All arithmetic bounds below refer to binary input size, not to the number of normal discs represented by a height vector.

The native validator accepts only finite compact orientable manifolds with one torus boundary. The isolated sharp ball in [section 6](#) is checked by an independent Regina driver; the native performance fixtures attach copies of that ball to a solid-torus boundary. No ball input is passed through a validator that claims to accept only torus boundary.

3 Backward persistence and causal factorization

Lemma 3.1 (Relative-boundary commutation). *Two simultaneously legal formal bipyramid replacements with disjoint consumed incarnation sets commute. Their two orders produce canonically isomorphic marked triangulations, fixing all unaffected cells and respecting the two replacement boundaries.*

Proof. Replace both abstract fillings before imposing their attaching maps. Their interiors are disjoint, and their individually labelled boundary triangles are unchanged. Gluing the two new fillings to the old exterior therefore gives the same quotient in either order. This includes a boundary triangle of one region paired to a boundary triangle of the other.

The upward event's internal paired face and the downward event's complete edge star of three tetrahedra are unaffected by a replacement with disjoint consumed cells. Hence each event remains legal after the other. The five formal height values are determined by the unchanged old edge differences; the resulting rows agree after the tracked relabelling and normalization. $\square$

The forward statement is insufficient to delete unrelated historical events: a move might have become enabled only after an earlier move. For the present model, such a dependency must involve an actual birth consumed by the later event.

Lemma 3.2 (Backward persistence). *Suppose $a; b$ is a legal adjacent pair of moves and b consumes no incarnation born at a . Then b was legal immediately before a , and $a; b$ can be interchanged to $b; a$ under the tracked boundary identifications.*

Proof. Every source incarnation of b existed before a and survived it. These cells are disjoint from those consumed by a . Pairings between two survivor tetrahedra, including their local permutations, are unchanged by a .

For an upward b , its internal face is exactly such a survivor-to-survivor pairing. Its two distinct tetrahedra therefore support the same upward move before a .

For a downward b , its three central local edge occurrences form a closed triangular edge-link cycle. Every face incident to these local edges is paired within the three survivor tetrahedra. Those same pairings existed before a . Edge equivalence is generated by the pairings of faces containing the local edge, so its orbit already closed within those three occurrences; it could not have included a fourth exterior occurrence. The formal belt and apex consistency checks involve only these unchanged permutations. Thus the entire downward legality condition already held. Apply lemma 3.1 to interchange the now simultaneously legal moves. $\square$

Definition 3.3 (Birth graph). The birth graph D_σ of a finite trace has its move events as vertices. It has one directed arc $a \rightarrow b$ for each tetrahedron incarnation born at a and consumed at b . Parallel arcs are retained. The graph is acyclic, and a component means a weakly connected component.

Theorem 3.4 (Component projection). *The subsequence formed by any union of birth-graph components is a legal trace from the original source, with the original local frames transported through it. The full trace can be reordered into any order of its component blocks. If a component C has u_C upward and d_C downward events, then the full loss is $\sum_C (d_C - u_C)$.*

Proof. Preserve the internal order of the selected and unselected subsequences, and move all selected events to the beginning by adjacent swaps. At a swap $a; b$ with a unselected and b selected, there is no birth arc from a to b : otherwise their components would coincide. Thus b consumes no output of a , and lemma 3.2 permits the interchange. Repeating finitely many swaps makes the selected subsequence an initial legal trace. The same procedure orders all component blocks. Marked endpoint agreement comes from lemma 3.1; the loss formula is (3). $\square$

This theorem preserves reachability of the projected trace, not every global property of the full endpoint. Additivity of tetrahedron loss is essential in the application below.

4 Connected original supports and the incidence bound

For a component C , let F_C be the set of original tetrahedra consumed by its events. The sets F_C are disjoint, since an original incarnation is consumed at most once. The following lemma concerns connectivity in the *original* dual graph, not just in some later triangulation.

Lemma 4.1 (Original footprint connectivity). *For every nonempty birth-graph component C , the induced graph $G_T[F_C]$ is connected.*

Proof. Maintain a partition of the original tetrahedra. Initially every block is a singleton and labels its corresponding current incarnation. When a move consumes incarnations labelled by blocks $A_1, \dots, A_s$, merge these blocks into $A = A_1 \cup \dots \cup A_s$. Relabel every surviving incarnation of an affected block by A , and label all new incarnations by A .

Two invariants are maintained. First, each block induces a connected subgraph of G_T . Second, between distinct blocks, current cut-face pairings are in a boundary-port-preserving bijection with the original face pairings crossing those blocks. Both assertions hold at the start.

The consumed tetrahedra of either move are connected in the current dual graph: an upward move consumes an adjacent pair, and a downward move consumes the triangular cycle around its central edge. Their distinct block labels therefore form a connected graph. By the cut-face invariant, every edge joining two such labels witnesses an original dual edge between the corresponding original blocks. Their union is connected. The replacement retains all six formal boundary triangles and their attaching permutations; every pairing to an unaffected block keeps its original boundary port. Pairings newly internal to the merged block cease to cross the partition. This proves preservation of both invariants.

An activated block records exactly the original ancestors of a current weak birth component. Consuming a born incarnation attaches its producing event component; consuming births from several blocks merges those components; an untouched original cell enters a block exactly at its first consumption. Consequently the final activated blocks are precisely the F_C . The first invariant proves the lemma. $\square$

Lemma 4.2 (Incidence bound). *A birth component with u upward and d downward events consumes at most $u + 2d + 1$ original tetrahedra. More generally, a trace with c birth components satisfies*

$$|F(\sigma)| \leq u + 2d + c. \quad (4)$$

Proof. Let q count consumptions of born incarnations. Each consumed cell is either original or born, and the move input sizes give the exact identity

$$|F(\sigma)| + q = 2u + 3d.$$

The birth multigraph has $u + d$ vertices and c weak components, hence at least $u + d - c$ arcs. Its arc count is exactly q , including parallel arcs. Subtract this lower bound from the identity. The connected case has $c = 1$. $\square$

5 Automatic localization of a strict descent

Theorem 5.1 (Causal localization). *If T admits any descending legal trace with at most U total upward moves, it admits a trace τ such that, for some $u \leq U$:*

$$u(\tau) = u, \quad d(\tau) = u + 1, \quad |\tau| = 2u + 1, \quad (5)$$

$$|T_\tau| = t - 1, \quad |F(\tau)| \leq 3u + 3 \leq 3U + 3. \quad (6)$$

The birth graph of τ is connected, and so is its actual original footprint in G_T .

Proof. Choose a minimum-length descending trace among those using at most U upward moves. No earlier prefix descends. Every event changes the count by exactly one, so the final count is exactly $t - 1$, giving $d = u + 1$.

If the birth graph had several components, their losses would sum to one. Some component would have positive loss. Its projection is legal by [theorem 3.4](#), uses no more upward moves,

and is a strictly shorter descending trace. This contradicts minimality. Hence the graph is connected. By lemma 4.1, its actual original footprint is connected; by lemma 4.2,

$$|F(\tau)| \leq u + 2(u + 1) + 1 = 3u + 3.$$

The length and remaining claims follow. $\square$

Remark 5.2 (Why one truncation is insufficient). Extracting a descending component and truncating it at first descent does not by itself prove that the retained birth graph remains connected. Later events may have merged components that were previously separate. The minimum-length argument obtains both properties at once. The implemented certificate extractor instead alternates component projection and first-descent truncation until stable; every nontrivial change removes an event.

Corollary 5.3 (Constructive certificate localization). *Given an explicit verified descending trace, a connected first-descending subtrace satisfying (5)–(6) can be found and replayed in time polynomial in the supplied trace’s bit length.*

Proof. Construct the birth graph by tracking live incarnations. Select a positive-loss weak component and project to it, then truncate at its first strict descent. Repeat. Each iteration retains a positive-loss trace and either terminates with a connected first descent or removes at least one event. There are at most as many iterations as input events, and each graph operation is polynomial. Replay the retained events with their original local frames. The projection theorem proves legality, and independent move and transport checkers verify the emitted certificate. The incidence calculation then applies. $\square$

6 A sharp six-tetrahedron example

Consider the simplicial ball with seven distinct vertices A, B, C, D, E, X, Y and six tetrahedra

$$ABCD, EBCD, ABCX, EBCX, ACDY, ECDY, \quad (7)$$

gluing equally named triangular faces. Start with the bipyramid $ABCD \cup EBCD$, then attach $ABCX \cup EBCX$ and $ACDY \cup ECDY$ along their respective two-triangle boundary discs. Each attachment is a ball attached along a disc, so the result is a 3-ball.

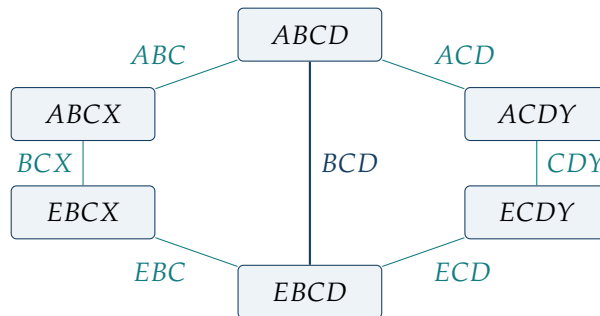


Figure 1: The original dual graph of the sharp ball. Each labelled edge is an internal paired triangle. The only internal primal edges are BC and CD , both of degree four. The central upward move enables two downward moves, whose union consumes every original tetrahedron.

Proposition 6.1 (Sharpness at one upward move). *The ball (7) has no initial downward move. It admits a descent using one upward and two downward moves, but every descent using at most one upward move consumes all six original tetrahedra. Thus the value $3U + 3 = 6$ cannot be reduced to five at $U = 1$ in the stated move model.*

Proof. The only internal edges are BC and CD , each with four incident tetrahedra. Therefore no downward move is initially legal. An upward move through BCD replaces the central two tetrahedra by $AEBC, AECD, AEDB$. The edge BC now supports a downward move using $AEBC, ABCX, EBCX$, producing $AEBX, AECX$. The edge CD supports the second downward move using $AECD, ACDY, ECDY$, producing $AECY, AEDY$. Five tetrahedra remain. The two downward moves commute in this displayed construction.

For the lower bound, a first descent with at most one upward move must have the pattern up/down/down. The upward move creates one new internal edge. If the first downward move deletes that edge, it undoes the upward move and returns to a triangulation with no downward move. Otherwise it deletes BC or CD . To reduce that edge's degree from four to three, the upward move must have used it in its central triangle. Its ensuing downward move consumes exactly one of the three new tetrahedra and produces two tetrahedra containing the newly created edge. That new edge's degree therefore changes from three to four and it cannot be deleted by the second downward move. The second move must delete the other of BC, CD .

An untouched original tetrahedron incident to an edge prevents that edge from disappearing. Hence deleting both edges forces consumption of the union of their original stars, which is precisely all six tetrahedra. $\square$

The accompanying independent Regina inventory examines every first upward move: seven candidates, fifteen up/down prefixes, and four successful up/down/down traces. Their first faces are BCD (two orders), BCX , and CDY ; all have footprint six. The initial and final isomorphism signatures are respectively `gbbjrcadfaa` and `fbbfiaev`. These finite records check the construction and the proof's edge inventory; they are not used to infer an asymptotic statement from a few examples. The article claims sharpness at $U = 1$, not asymptotic optimality of the coefficient three for all U .

7 A category boundary: strict simplicial moves

The hypotheses about generalized triangulations cannot be omitted. Let S be the boundary of an octahedron and form its simplicial suspension with vertices x, y . This triangulates a 3-sphere. For each base triangle abc , the tetrahedra $xabc, yabc$ admit a strict simplicial 2–3 move because the proposed edge xy is initially absent.

Choose two opposite base triangles. Their two move supports are disjoint, and both moves are initially legal. Performing either creates the edge xy , so the other is then forbidden by the strict simplicial nonexistence condition for the new simplex. Disjoint consumed tetrahedra therefore do not imply commutation in that category. Reversing the first move gives a backward-persistence failure caused by deletion of a global obstruction, without consumption of any tetrahedron used by the newly enabled move.

The repository's generalized model allows distinct edges with the same global endpoints and has no such absent-simplex condition. Extending the theorem to a different move model requires recording its additional dependencies and reproving the locality bound.

8 Complete fixed-parameter simplification

8.1 Constructively enumerating connected regions

For completeness, we give the combinatorial ingredients of the incoming region bound explicitly. This separates the new localization theorem from the search machinery on which its stronger algorithmic consequence rests.

Lemma 8.1 (Connected-region count). *In a graph with t vertices and maximum degree four, the number of connected i -vertex sets is at most $t16^{i-1}$ for $i \geq 1$. All connected sets of size at most R can be enumerated without repetition using polynomial work per generated set.*

Proof. Choose the least vertex as root and a deterministic spanning tree of each connected set. A depth-first traversal of the tree has $2(i-1)$ edge steps. Record the root and, at each step, one of at most four local neighbour ports. The walk reconstructs its visited vertex set, so this is an injective encoding among the deterministic choices. There are at most $t4^{2(i-1)}$ such encodings.

To enumerate sets, define the parent of a non-singleton connected set A by deleting its largest non-root vertex whose deletion leaves a connected set. A spanning tree has a leaf other than the root, so at least one deletion exists. Iterating the parent reaches the singleton root. From each current set, consider its at most $4|A|$ frontier neighbours and accept an extension exactly when its canonical parent is the current set. This visits each set once. Testing all candidate deletions and their connectivity uses only polynomial work in $|A|$. An explicit stack gives the implementation without recursion-depth dependence on the ambient number of vertices. $\square$

The geometric factor t chooses the location of the region. It is not raised to a power proportional to U . This is the step that direct enumeration of all R -subsets would lose.

8.2 An elementary independent FPT bound

Before using commutation compression, localization alone already implies fixed-parameter tractability. In a region containing i original cells, at most $i + U$ eligible cells are live at any time. A legal eligible event is specified by a local face or edge of such a cell; there are at most $10(i + U)$ choices before eliminating repetitions and illegal configurations. Every first-descent witness has at most $2U + 1$ events. Thus direct local branching, followed over every connected region of size at most $3U + 3$, gives

$$2^{O(U \log(U+2))} \text{poly}(t + B). \quad (8)$$

This proof supplies a separate, weaker FPT guarantee that does not require the constant-alphabet trace theorem. The sharper result below removes the extra U in the exponent.

8.3 Constant-alphabet encoding of commuting traces

Assign each eligible incarnation a commitment from

$$\mathcal{A} = \{\text{idle}\} \cup \{\text{down at edge } e : 0 \leq e < 6\} \cup \{\text{up at face } f : 0 \leq f < 4\}. \quad (9)$$

An event is ready if all its participants choose their local port for that event. Two different ready events cannot share an incarnation: a local face determines its unique paired-face event,

and a local edge determines its unique full edge-star event. Ready events therefore commute by lemma 3.1.

Fix a deterministic scheduler. If an event is ready, execute the first ready event; otherwise assign the next unassigned eligible incarnation one of its eleven symbols. Surviving cells retain their commitments; born cells are unassigned. Protected original cells are permanently idle.

Lemma 8.2 (Commitment encoding). *For a supplied region with i original cells, every finite trace using u upward and d downward events has an encoding of length $i + 3u + 2d$ over (9), up to interchanges of independent events. Consequently the number of commuting classes with $u \leq U$ and $d \leq u + 1$ is at most*

$$11^{i+5U+2}. \quad (10)$$

The generic bounded-region family, without the latter loss restriction, has at most 11^{3i+5U} classes.

Proof. For a target trace, give each incarnation the port of the event that eventually consumes it, and give every target survivor the symbol idle. Follow these choices when the deterministic scheduler asks for them.

If an event is ready, consider the first remaining target event touching one of its source incarnations. All earlier events leave those sources intact. Its commitment identifies exactly that ready event, so the first touch must be the ready event itself. All preceding target events have disjoint consumed supports; the persistence and relative-boundary lemmas move the ready event to the front. Iterating realizes the target trace up to commuting independent events. Assigning the recorded symbols cannot stop before the next target event is ready, and target survivors are idle.

The freely assigned incarnations comprise the i eligible originals and the $3u + 2d$ born cells. Each receives one symbol. Record assignments in the order requested by the deterministic scheduler. Padding by idle symbols to a common maximum length preserves an injective encoding of trace classes: decoding a word determines one execution. Substituting $d \leq u + 1$ yields (10). In a general region-supported trace, protected originals survive, so $d \leq i + u$, giving the final bound. $\square$

The counting decoder in this lemma is allowed to pass an earlier descent introduced by reordering. It need not follow the implementation's early-stop policy. The target trace's event counts and total encoding length are what establish the injection. Confusing these two roles would unnecessarily weaken or invalidate the counting argument.

The practical search uses sleep sets instead of literal assignments. It explores enabled events in deterministic order and remembers an earlier sibling event across a later branch only while its consumed incarnation set is disjoint from each intervening event. Once a source incarnation is destroyed it never reappears. This nonresurrection, local event uniqueness, and persistence imply that at most one representative of a commuting class is visited. Conversely, the first representative of each class in the fixed search order cannot be blocked: a blocking event could be commuted left to obtain an earlier representative. This is the exact finite sleep-set argument specialized to these geometric events, as implemented in the incoming engine [2].

Theorem 8.3 (Complete bounded-upward descent). *For a valid generalized source T of size $t \geq 1$, an optional integral marking of input bit bound B , and $U \geq 0$, question (1) is decidable in*

$$t 2^{O(U)} \text{poly}(t + U + B) = 2^{O(U)} \text{poly}(t + B) \quad (11)$$

bit operations. A positive answer supplies a geometrically replayable first-descent trace. No spatial-coverage hypothesis is required.

Proof. Set $R = \min(t, 3U + 3)$. Enumerate all connected original regions of size at most R and search their bounded-upward traces with the endpoint predicate $|T'| < t$. A positive test returns immediately. If no positive is found, every permitted family is exhausted. The localization theorem proves that this decides (1) completely.

Every visited prefix before success has $d \leq u$, and the first successful one has $d = u + 1$. Combining lemmas 8.1 and 8.2, a sufficient counting factor, including the source endpoint, is

$$1 + t \sum_{i=1}^R 16^{i-1} 11^{i+5U+2}. \quad (12)$$

Since $R \leq 3U + 3$, this is $t2^{O(U)}$ times a fixed constant. In particular, the largest commitment exponent is $8U + 5$, improving the $14U + 9$ obtained by substituting R into the generic region bound. The bit analysis below supplies polynomial work per visited encoding or representative. It includes the endpoint count test. This proves (11). $\square$

Corollary 8.4 (Small-budget regimes). *Bounded-upward descent is polynomial-time when $U = O(\log(t + 2))$, and takes $t^{O(\log(t+2))} \text{poly}(B)$ time when $U = O(\log^2(t + 2))$.*

Proof. Substitute the budget into (11). Polynomial factors in U are absorbed by the indicated bounds. The bit bound B remains polynomially charged. $\square$

The constants in (12) are intentionally conservative. It is an asymptotic guarantee, not a prediction that large U is practical. For $U = 0$, the problem is simply the existence of a legal initial downward move, since any descending trace has a first downward move. Directly checking edge stars solves that special case without a general region enumeration.

8.4 Binary arithmetic and certificate size

Lemma 8.5 (Upward moves control marking growth). *Let D_0 be the maximum absolute source edge value. After a trace with u upward events, all signed edge values have magnitude at most $2^u D_0$. Normalized height rows and local coherent normal coordinates have bit length at most $B + u + O(1)$.*

Proof. A downward move removes one edge and creates no new edge; all surviving edge values remain unchanged. The new diagonal of an upward move joins the two formal apices a, b . Through any belt vertex v , its value is $(h_b - h_v) + (h_v - h_a)$, a sum of two old signed edge values. Its magnitude is at most twice the previous maximum, and all other edge values persist. Induct on u . A normalized row entry is an edge difference from its first corner. A local normal coordinate is a gap between ordered corner heights, bounded by their span, which is itself an edge difference. $\square$

There are $O(t + U)$ local edge and face occurrences at every visited state. Global manifold validation, grouping edge stars, comparing local patterns, and copying a full triangulation therefore cost polynomial time in $t + U + B$. The code currently rebuilds global geometry where needed; it makes no constant-time local-update claim. Stable incarnation names are exact interned DAG expressions, not cryptographic identity assumptions.

A first-descent proof with complete intermediate snapshots has bit length

$$O((U + 1)(t + U)(B + U + \log(t + U + 1))). \quad (13)$$

It contains at most $2U + 1$ moves, each with at most $t + U$ rows, plus the final coordinate data. Independent replay is polynomial in this size. Exhaustive failure is an algorithmic conclusion about a bounded family; the implementation does not claim a comparably short negative certificate.

8.5 Repeated bounded simplification

Definition 8.6. A triangulation is U -irreducible for this move model if it admits no strict descent using at most U total upward moves.

Corollary 8.7 (Complete bounded preprocessing). *Repeatedly applying complete bounded- U descent and adopting each verified positive result terminates at a U -irreducible triangulation in $2^{O(U)} \text{poly}(t + B)$ bit time. Its concatenated positive proof uses at most tU upward events and $t(2U + 1)$ events in total.*

Proof. Each successful restart reduces the tetrahedron count by at least one, so there are fewer than t successful restarts. Each local transient state has at most $t + U$ tetrahedra. Across all successful traces, [lemma 8.5](#) increases the marking bit bound by at most $tU + O(1)$. Summing the polynomial overheads in [theorem 8.3](#) gives the stated bound. The final complete negative bounded search is precisely the definition of U -irreducibility. $\square$

An arbitrary choice of successful one-unit descents is not guaranteed to preserve a prescribed total upward budget for a later k -unit objective. The next section treats that joint-budget problem separately.

9 Several units of descent and color coding

The connected witness theorem answers one-unit descent. Requiring a decrease by at least k tetrahedra may require several independent regions. Their locations need not be enumerated with a t^k factor: their total original footprint remains bounded, which permits standard deterministic color coding. This section gives a further algorithmic theorem. The delivered code includes the exact packing primitive for a supplied coloring and finite audits; it does not implement the full deterministic perfect-hash-family construction.

Theorem 9.1 (Multiunit localization). *Suppose a trace with at most U upward events reduces T by at least $k \geq 1$ tetrahedra. Then there is such a trace with $u \leq U$, $d = u + k$, length $2u + k$, and at most k birth components. Each component has positive loss and connected actual original footprint. Their disjoint union has size at most $3U + 3k$.*

Proof. Choose a shortest trace meeting the threshold. It first reaches loss k at its final event, so $d = u + k$. If a component had nonpositive loss, deleting it using [theorem 3.4](#) would preserve loss at least k and shorten the trace. All component losses ℓ_j are therefore positive integers, with $\sum_j \ell_j = k$, so the number c of components is at most k . Their footprints are disjoint and connected by [lemma 4.1](#). Summing [lemma 4.2](#) yields

$$|F| \leq u + 2d + c = 3u + 2k + c \leq 3U + 3k.$$

$\square$

Lemma 9.2 (Composing disjoint original footprints). *Let $\sigma_1, \dots, \sigma_s$ be legal traces from the same marked source whose actual original footprints are pairwise disjoint. They can be replayed in any order, transporting the exterior attachments at each step. The combined upward cost and tetrahedron loss are the sums of the individual costs and losses.*

Proof. Give births in different traces disjoint names. Every incarnation consumed by one trace is either one of its original footprint cells or a birth descended from those cells. Distinct traces

therefore have disjoint consumed incarnation sets after transport. At the common source their first events are simultaneously legal and commute. Inductively construct the grid of interleavings: at every square, [lemma 3.1](#) transports the next event of either trace across the next event of the other, retaining the same relative-boundary attachments. This proves that both complete orders are legal. Apply the two-trace argument repeatedly. Marking agreement follows from the same local transport lemma, and the count assertions follow by adding event counts. $\square$

This lemma does not require an actual footprint to be a ball or an induced subcomplex with embedded closure. Individual formal replacements, with their boundary ports, are the units of the proof. It also does not claim that arbitrary traces with overlapping original footprints compose.

9.1 Enumerating candidate items

Set $p = \min(t, 3U + 3k)$. Over every connected original cover of size at most p , enumerate the full bounded-upward region family and retain every positive-loss endpoint. Each item j records

$$(A_j, a_j, b_j, \sigma_j),$$

Here A_j is its *actual* consumed original footprint, $a_j \leq U$ is its upward cost, and σ_j is a replayable source-bound trace. The positive profit is capped: $b_j = \min(k, \text{loss}(\sigma_j))$. The total item count and generation cost are bounded by

$$M \leq t 2^{O(p+U)} \text{poly}(t + U + B). \quad (14)$$

This is the generic connected-region commitment bound. A particular actual footprint can appear under several covers; retaining duplicate items does not affect the upper bound or correctness.

The item generator must continue through intermediate descents. Calling a first-descent routine and collecting only its first answer would not enumerate the needed component alternatives. Birth-connectedness need not be tested for every retained item: extra legal items remain sound, while every component of the minimum witness in [theorem 9.1](#) is necessarily represented.

9.2 Colorful packing with original witnesses

A family $\mathcal{H}$ of functions $[t] \rightarrow [p]$ is p -perfect if every subset of at most p source indices is mapped injectively by at least one member. The standard deterministic construction of Naor, Schulman, and Srinivasan has size $e^p p^{O(\log p)} \log(t+2)$ and total construction time bounded by $2^{O(p)} \text{poly}(t)$ [9]. We need this total-time statement; we do not assume a constant-delay hash enumerator. Color coding itself is classical [10]; an explicit modern treatment of the stronger delay requirements is available in [11].

For a fixed coloring $h \in \mathcal{H}$, discard an item if h is not injective on A_j . Otherwise store its nonempty color mask $C_j = h(A_j) \subseteq [p]$. Use the following ordinary 0/1 dynamic program:

$$\begin{aligned} D_0(\emptyset, 0) &= 0, & D_0(S, a) &= -\infty \text{ otherwise,} \\ D_j(S, a) &\leftarrow D_{j-1}(S, a), \\ D_j(S \cup C_j, a + a_j) &\leftarrow \max\{D_j(S \cup C_j, a + a_j), \min(k, D_{j-1}(S, a) + b_j)\} \end{aligned}$$

for every finite state with $S \cap C_j = \emptyset$ and $a + a_j \leq U$. The state value is the best capped total loss from the first j items, using exactly mask S and upward cost a . Take/skip predecessor data retains actual item identities for reconstruction; a linear combination of traces is never treated as a geometric witness.

The number of states is $2^p(U+1)$, and a direct item scan costs $O(M2^p(U+1))$. Positive items have nonempty footprints, so the empty-mask degeneracy does not arise. Mask disjointness implies actual source-cell disjointness, even though distinct source cells may share colors. By lemma 9.2, a successful collection is geometrically replayable and has total upward cost at most U and loss at least k .

Theorem 9.3 (Fixed-parameter multiunit descent). *For generalized formal-bipyramid moves, existence of a trace with at most U total upward moves reducing a valid t -tetrahedron source by at least k tetrahedra is decidable deterministically in*

$$2^{O(U+k)} \text{poly}(t+B) \quad (15)$$

bit time. A positive result admits a source-bound geometric trace certificate.

Proof. Trivial impossible thresholds, including $k \geq t$ for a nonempty connected source, may be answered directly. In the remaining cases construct the items above and run the packing dynamic program for every coloring in $\mathcal{H}$. Soundness follows from lemma 9.2.

For completeness, take the minimum-length witness of theorem 9.1. Its component footprints are individually connected, pairwise disjoint, and their union has at most p vertices. Each component trace occurs among the items. Some $h \in \mathcal{H}$ colors the entire union injectively, so these items have pairwise disjoint color masks. Their upward costs sum to at most U and their profits sum to k ; the dynamic program finds a successful collection. Enumerating every coloring therefore decides the question completely.

Multiply (14), the $2^p(U+1)$ dynamic-program factor, and the $2^{O(p)} \log(t+2)$ hash-family factor. Since $p \leq 3(U+k)$, the result is (15), with polynomial source and replay costs included. $\square$

The weaker direct alternative is to enumerate covers with up to k components, which gives $t^k 2^{O(U+k)} \text{poly}(t+U+B)$. The color-coding argument removes the t^k factor. This is a consequence of the new bounded total footprint and independent replay, composed with an established derandomization theorem. It is not a claim that color coding is new.

With $U+k = O(\log t)$ the decision problem is polynomial; with $U+k = O(\log^2 t)$ it is quasi-polynomial. The supplied-coloring prototype tests the witness-preserving packing step, while theorem 9.3 relies on the cited complete perfect-family construction for deterministic exhaustion over all possible locations.

10 One exact search over overlapping covers

The localization theorem makes the old spatial search complete for descent, but independent restarts still repeat substantial work. A source with many small overlapping covers can revisit the same marked trace prefix once for each cover containing its footprint. The new implementation represents the union of those finite search languages directly.

10.1 The cover intersection invariant

Let $\mathcal{R}$ be the complete family of permitted original regions, including the empty region. For each original tetrahedron v , build the inverted incidence list

$$I(v) = \{R \in \mathcal{R} : v \in R\}.$$

A search frame with actual footprint F retains precisely

$$C(F) = \{R \in \mathcal{R} : F \subseteq R\} = \bigcap_{v \in F} I(v). \quad (16)$$

The empty intersection means all covers. The implementation represents that root case by a sentinel instead of copying the complete index. If an event first consumes the original cells in S , its compatible covers become

$$C(F \cup S) = C(F) \cap \bigcap_{v \in S} I(v).$$

An empty result rejects that branch. Consuming a born cell adds no new original identity: its ancestors already entered the footprint at their own first consumption. Intersections begin with the smallest available incidence set, reducing intermediate allocations without changing the set.

Proposition 10.1 (Exact union language). *The indexed cover search permits exactly the traces admitted by at least one independently restarted search over $\mathcal{R}$. It preserves the relative commutation equivalence used by the sleep-set search.*

Proof. Induction on the trace length proves (16). A trace is admitted exactly when some region contains every consumed original cell, which is the definition of membership in the union of the restarted languages. Every prefix consumes a subset of that final footprint, so it is admitted under the same covering region.

For two adjacent independent events in an admitted trace, either order has the same final footprint, while the two intermediate footprints are subsets of it. The same cover therefore admits both orders. Their geometry and markings agree by [lemma 3.1](#). Thus all swaps required by the sleep-set argument remain within the union language. $\square$

Two events may be individually allowed at a prefix although their combined footprint fits no cover. The proposition does not assert that every such pair can be combined. It asserts commutation when the whole two-event trace is admitted, which is exactly the property the reduction needs.

An explicit index has at most $\sum_{R \in \mathcal{R}} |R|$ incidences. For connected radius $R_0 = O(U)$ this is $t2^{O(U)}$ times a polynomial parameter factor. Intersecting finite sets and copying compatible-cover states keeps the same fixed-parameter time form, although it can consume more memory than streaming one region at a time. The theorem does not promise that a fully materialized index is the best implementation for every input.

There is also a sharper accounting for cover intersections. Let C_f be the compatible-cover set at visited frame f , and let K_R bound the commuting trace classes admitted by a fixed cover R . Then

$$\sum_f |C_f| = \sum_{R \in \mathcal{R}} \#\{f : R \in C_f\} \leq \sum_{R \in \mathcal{R}} K_R.$$

For a fixed cover, the global sleep search still visits at most one representative of each admitted class. An intersection can scan the smaller set; ordered incidence tables with binary search

give a deterministic logarithmic overhead per tested membership. Thus incidence work can be charged to compatible cover/frame pairs instead of multiplying every frame by the size of the entire region family. The prototype uses exact Python sets, with equality resolving hash collisions.

10.2 Lazy transitions and endpoint observations

Each queued branch stores its parent state and the proposed geometric event. The native replacement and integral cochain transport are performed only when that branch is visited. A positive early answer can therefore avoid constructing all unused siblings. The node limit is checked before producing the next unvisited child. This matters on the upward-required fixtures: the source has many upward choices, while a successful trace uses only three events.

Exact marked endpoint keys suppress repeated endpoint queries. They retain the source incarnation identities and transported cochain, using exact interned structures rather than probabilistic hash equality. The search does *not* delete continuation frames merely because an endpoint query was already made. Remaining upward budget, sleep obligations, and compatible covers can differ between histories reaching similar geometry. Proving a safe stronger quotient is a separate problem.

A complete conclusion for a caller's predicate requires that predicate to be invariant under the represented marked endpoint and commuting trace equivalence. A callback that inspects the particular certificate history or arbitrarily selected witness cover is outside that guarantee. Strict tetrahedron descent and the supplied-vector disc predicate have the required invariance.

The finite-family cost separates three tasks: construction of the complete cover index, exploration of geometric traces, and evaluation of endpoints. The descent endpoint test is just a tetrahedron count. A normal-disc query is optional and has its own potentially substantial cost. Performance claims for the descent test therefore do not automatically transfer to an arbitrary disc-search callback.

10.3 Public interfaces and exact meanings of results

The generic entry point is `search_pachner_cover` in `fastunknot/pachner_cover_search.py`. It accepts a source triangulation and height rows, a maximum region size, maximum number of components, total upward budget, and explicit resource limits. The methods are the maintained sleep-set route and a small-instance naive reference route. The older restarted and commitment implementations remain available.

The convenience entry point `find_pachner_descent` supplies the radius $\min(t, 3U + 3)$ unless the caller explicitly requests another radius. It queries strict tetrahedron loss and independently verifies a positive answer. A smaller requested radius is a legitimate local search, but its failure has a deliberately narrower status.

The generic interface also distinguishes an independently verified normal-disc answer, `DISC_FOUND`, from a caller-selected endpoint, `ENDPOINT_SELECTED`. A certificate on an abstract manifold still needs the maintained diagram-to-exterior binding before it can certify a statement about a supplied knot diagram.

10.4 Independent certificate checking

The producer emits the existing `pachner-cochain-endpoint-v1` schema. Its declared active original region is one concrete member of the compatible-cover set; it need not equal the actual

Status	Meaning
DESCENT_FOUND	Source replay verifies strict descent and the declared total upward bound.
COMPLETE_BOUNDED_DESCENT	No descent exists within that upward budget, using the sufficient localization radius and complete search.
COMPLETE_BOUNDED_LOCAL_DESCENT	Only the explicitly smaller covered family is exhausted.
COMPLETE_BOUNDED_COVER_FAMILY	The complete declared generic cover family is exhausted.
INCONCLUSIVE	A required index, traversal, replay, or endpoint-query operation reached its limit.

Table 1: The principal search outcomes. Exhaustion is a bounded geometric conclusion and is not a nontrivial-knot certificate.

Listing 1: Complete bounded descent with an independently checked positive witness.

```

from fastunknot.pachner_cover_search import find_pachner_descent
from fastunknot.pachner_cover_verify import verify_pachner_descent

answer = find_pachner_descent(
    triangulation, heights, max_upward=1,
    max_nodes=10000, max_regions=100000, max_work=2000000,
)
if answer["status"] == "DESCENT_FOUND":
    assert verify_pachner_descent(
        triangulation, heights, answer["certificate"],
        max_upward=1,
    )

```

consumed footprint. The new verifier independently checks that this cover has the stated size and number of induced dual components, then invokes the maintained source-bound geometric and cochain replay. It imports no region enumerator, search engine, native move producer, or cochain-transport producer.

For descent, it also recomputes upward and downward counts, checks the actual final tetrahedron count, and verifies $t_{\text{final}} = t_{\text{initial}} + u - d < t_{\text{initial}}$. This prevents the search's status string or claimed counters from substituting for a proof. The verification tests disable the corresponding producer functions and still accept genuine retained certificates.

Resource caps are part of the observable contract. `max_regions` limits construction of the *complete* index, including the empty region. Partial construction cannot establish complete exhaustion. `max_nodes` counts visited search frames; `max_work` counts cooperative checkpoints across preparation, indexing, search, and positive replay. These checkpoints are operational accounting units, not bit operations. Optional disc-query caps also prevent a complete disc-negative claim when a required query was unfinished. User callback exceptions propagate unchanged, including exceptions whose class happens to match an internal budget exception. Callback inputs are isolated from mutable search state.

11 Certificate projection, composition, and packing

11.1 Extracting a connected first descent

The new `pachner_causality.py` reads a supplied endpoint proof only after independent source replay accepts it. Each live tetrahedron receives either a source identity or a pair consisting of its birth event and output slot. Reading the consumed regions recovers every birth arc and the actual source footprint. The analyzer reports the components, their losses, and the incidence census.

For a descending certificate, `extract_local_descent` chooses a positive-loss component, projects its events, and truncates at first descent. These operations repeat until the retained trace is connected and first-descending. One projection followed by one truncation is insufficient: deleting a late event that joined earlier components can disconnect the prefix again. Every necessary repetition strictly shortens the trace, giving termination. The final witness satisfies the localization bound and is independently replayed before return.

The consumer must respect every local frame accepted by the public verifier. A valid downward certificate can use either order for two belt vertices in the seed tetrahedron. A native producer may choose just one order. Replaying with that producer's frame and then treating its born tetrahedra as if they used the declared frame corrupts later dependent events, even though the first replacement has the right unmarked geometry. The extractor therefore retains the declared formal region and, where necessary, relabels the two born tetrahedra and all incident permutations before carrying the cochain. This fix is exercised by a retained noncanonical-frame counterexample and an exhaustive finite frame audit described in [section 12](#).

11.2 Composing independently supplied traces

The companion `compose_disjoint_traces` authenticates every input endpoint against the same original source. It computes *actual* consumed footprints and rejects intersecting ones, even if their advertised cover regions happen to differ. Birth names are namespaced by input trace, so equal event numbers from different certificates cannot collide. The replay transports source and birth identities, preserving declared formal frames, and returns one ordinary source-bound endpoint certificate together with the additive count census. This is an executable positive-witness counterpart of [lemma 9.2](#). Overlapping supplied covers are harmless when the actual footprints are disjoint.

The function does not assume that input traces are descending. The geometric composition lemma applies to arbitrary legal traces with disjoint actual footprints; a caller may separately inspect their losses. Empty traces are allowed and contribute zero. Invalid proofs, overlap, or exhausted work do not produce a partial successful composition.

11.3 The supplied-coloring dynamic program

The prototype `causal_research/colorful_packing.py` implements the exact 0/1 packing step of [section 9](#) for a supplied coloring. It validates nonempty distinct-source footprints, non-negative upward costs, and positive losses. It discards noncolorful items, compacts unused numeric color labels, and preserves original candidate indices in immutable witness tuples. A separate previous-item table prevents accidental reuse of an item.

Its objective is maximum capped loss, followed by minimum upward cost. It does not claim globally minimum witness cardinality or lexicographically first item indices: retaining only

one best capped profit per mask and cost does not justify those extra objectives. The result exposes actual loss as well as capped loss, upward cost, union footprint, and color mask. The candidate dictionaries can retain authenticated certificates, but the combinatorial routine itself does not authenticate their geometry.

The delivered prototype does not generate the complete deterministic perfect-hash family. Consequently a negative result from one coloring is only a negative result for that colorful item instance. The mathematical algorithm in [theorem 9.3](#), the implemented packing primitive, and the independent positive composition utility are separate, explicit deliverables.

12 Experimental evidence and its limits

The experiments test equivalence of bounded searches, independent replay, the default localization radius, and the cost of repeated overlap. They do not measure a complete knot-recognition pipeline. All raw data, source digests, generated fixtures, and replay commands are retained.

12.1 Genuine upward-required solid-torus fixtures

Start with a one-tetrahedron layered solid torus and attach a chain of g copies of the six-tetrahedron ball along boundary triangles. Attaching a ball along a boundary disc preserves the solid torus. The resulting source has $t = 6g + 1$ tetrahedra. Its cocycle is computed and validated using the maintained exact native routines. Independent Regina checks confirm validity, orientability, and the solid-torus type for the controlled small instances [\[12\]](#).

These sources have no initial downward move; the native move inventory has 8g upward choices. A local first descent requires one upward and two downward moves and consumes six original gadget cells. For the one-gadget source the complete $U = 0$ test fails to descend, and the complete $U = 1, r = 5$ covered search also fails. With $U = 1, r = 6$, the native search returns an independently accepted descent. This distinguishes a genuinely upward-required test from an example solved by an initial greedy collapse. The separate ball proof and Regina inventory establish the sharper radius statement of [proposition 6.1](#).

12.2 Matched timing protocol

The retained measurements use Python 3.12.14 on Linux-6.18.44-x86_64-with-glibc2.39. Every source is constructed once. The independent controls use Regina engine 7.4 (Python package 7.4.1). Both routes use the same $U = 1, r = 6$, cochain, native move and transport routines, and relevant resource limits. Disc queries are disabled. Each reported time is the median of three retained samples; the run order alternates between restarted and shared search. The speed ratio is the ratio of those two medians, not the median of per-pair ratios.

For complete-family timings, endpoint collection is disabled in both routes. Endpoint equality is checked separately. For first-descent timings, both routes include the same independent positive verifier. Source fixture construction is outside the timed calls, while region enumeration and index construction are inside. Timing uses a shared execution environment, without dedicated CPU isolation. Three samples document a reproducible finite comparison but do not support a population confidence interval or an extrapolated running-time law.

In the complete runs, cooperative work checkpoints fall from 367,010 to 23,374 at $t = 7$, and from 4,136,232 to 84,460 at $t = 13$. The changes are consistent with removing repeated common prefixes across overlapping covers. In the first-descent runs the shared method

t	Covers	Visited frames		Median seconds		Ratio	Endpoints
		Restart	Shared	Restart	Shared		
7	66	495	28	1.15956	0.09137	$12.69\times$	14
13	334	3152	55	13.39354	0.22377	$59.85\times$	27

Table 2: Complete covered-family exhaustion. “Endpoints” counts exact signed-cochain geometric isomorphism classes, checked outside the timed runs. The shared search retains more marked representatives.

t	Covers	Visited frames		Median seconds		Ratio
		Restart	Shared	Restart	Shared	
7	66	273	6	0.71148	0.01801	$39.50\times$
13	334	630	6	2.75100	0.03303	$83.29\times$
25	934	641	6	6.03841	0.10123	$59.65\times$
49	2134	641	6	10.68172	0.14371	$74.33\times$

Table 3: First independently verified strict descent on the controlled solid-torus family. Every selected witness uses one upward and two downward moves. These are simplifier measurements, not end-to-end unknot timings.

visits six frames and materializes five transitions; its selected proof has three events, so two materialized transitions belong to unsuccessful visited alternatives. Lazy materialization avoids paying for the unvisited siblings. Constant visited-frame counts do not imply constant runtime: the cover index and global geometry still grow with the source.

12.3 Exact endpoint equality and standalone replay

For $t = 7$ and $t = 13$, complete restarted and shared enumeration produce identical sets of full exact signed-cochain isomorphism tuples: respectively 14 and 27 classes. This comparison forgets source incarnation names but retains the geometric face-pairing structure and integral marking. The shared search retains 28 and 55 marked endpoints. Thus the audit does not claim that the two implementations visit identical histories or have a pointwise bijection between their internal frames.

Independent source-bound replay separately authenticates reachability and scope. The complete-family artifact’s standalone replay accepts 124 retained covered endpoint certificates and six strict-descent certificates. The first-descent artifact’s replay accepts another twelve strict descents. These counts include retained representatives and repeated timing witnesses according to the artifact schema; they are not numbers of distinct knot types or geometric states.

12.4 Adversarial and combinatorial audits

The targeted cover tests compare inverted-index intersections with literal subset containment, compare complete endpoint families against both older restarted and naive routes, and exercise independent replay with producers disabled. They distinguish actual support from a supplied connected cover, including a disconnected consumed subset. They test incomplete indexing, exact work boundaries, a one-node cap before transport, callback mutation isolation, and propagation of exceptions raised while converting a callback result to a truth value.

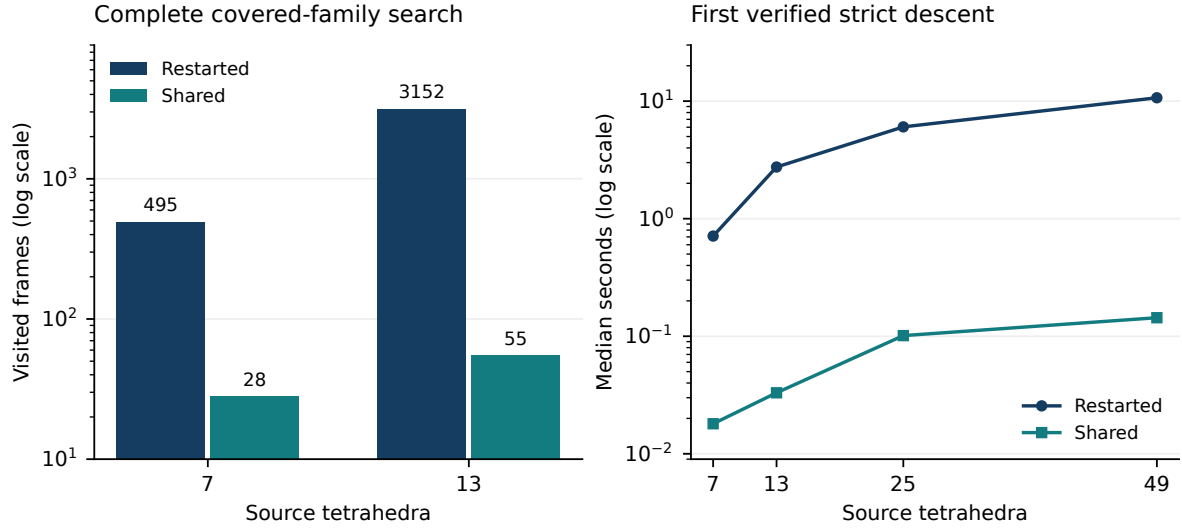


Figure 2: Two views of the same retained data. Logarithmic vertical axes make the smaller shared-search counts and times legible. Lines connect the four measured first-descent cases and do not represent a fitted asymptotic model.

The causal tests cover independent event components, positive projection, the late-merger case that requires repeated projection and truncation, invalid or non-descending inputs, and declared noncanonical downward frames. The portable frame audit constructs 144 formal variants on two dependent downward steps from fixed verified data and explicit cell/vertex renamings. Every input and extracted output passes independent replay; the exact final triangulation, normalized marking, and coherent coordinates agree in every case. In 48 variants the final seed is a child of the preceding downward move, directly exercising the frame-sensitive dependency. No search or Pachner producer constructs those input variants.

The colorful-packing tests compare every subset of 96 deterministic random item families under four budget/goal choices. They also test zero-cost items, empty inputs, invalid or repeated footprints, color-label compaction, discarded-item index preservation, and exact witness reconstruction after profit capping. On the genuine two-gadget source, all 55 retained endpoints are independently replayed and yield six positive candidates: three on each of the disjoint six-cell supports. One supplied coloring permits packing two items with total upward cost two and loss two. An exhaustive check of all 64 subsets finds nine optimal pairs and agrees with the DP.

The composition audit then replays a selected pair and produces a single certificate for $13 \rightarrow 11$ tetrahedra with $u = 2, d = 4$. Independent replay accepts both source proofs and the composed proof. Tests also reverse the order, allow overlapping declared covers with disjoint actual supports, reject overlapping actual supports and forged evidence, preserve noncanonical frames, and exercise empty composition and exact work limits. A further portable audit constructs twelve paired belt/apex frame variants of the second nonempty input trace and composes them after the first trace has changed row positions. Every composed endpoint verifies with the correct private birth namespaces and additive counts. These twelve focused composition cases are distinct from the exhaustive 144-frame extraction audit.

Three implementation issues found during development are retained as regression targets. An eager branch producer did unnecessary sibling transport before the next node-limit check; deferred materialization fixes this. A causal replay initially assumed the native producer’s downward frame was the only verified frame; preserving the certificate’s declared frame fixes

the dependent-step failure. The cover search also now shields the truth-value conversion of an endpoint callback result, so a callback exception cannot be mistaken for an internal work-limit signal. These corrections improve the trust boundary as well as measured performance.

12.5 Complete regression and reproducibility

The complete current inventory contains 1483 tests in 187 modules; all pass with zero failures, errors, or skips. Verification uses six fresh interpreter batches. After adapting one incoming test to the newer native unit-ray optimization, the affected fifth batch (315 tests) was rerun successfully. Source hashes confirm that the other tested modules and every production module were unchanged.

The initial runner required an explicit import path for inherited test helpers. Its failed setup record, the corrected run, the compatibility patch, and the successful affected-batch rerun are retained. The adapted test separately checks the native zero-orbit optimization and forces the documented fallback schedule to preserve the original exact-budget and under-budget assertions. No native production file was replaced or altered. The final machine-readable aggregate lists every test ID, its batch evidence, and current source hashes; it does not erase the earlier diagnostic runs.

Run the principal checks from the delivered repro/fast directory:

```
python -m unittest tests.test_pachner_cover_search \
    tests.test_pachner_causality tests.test_pachner_composition \
    tests.test_colorful_packing -v
python -m shared_cover_research.benchmark \
    --replay shared_cover_research/results/exhaustion.json
python -m shared_cover_research.benchmark \
    --replay shared_cover_research/results/descent.json
python -m causal_research.composition_audit \
    --replay causal_research/results/disjoint_composition.json \
    --output reproduced/composition_replay.json
```

The package-level regression runner discovers the complete delivered test inventory, records hashes and test IDs, and runs six fresh interpreters. Fresh output directories keep reproduction results separate from recorded evidence. The frame and ball audit drivers, paired benchmark generator, figure generator, and checksum verifier have their own documented commands.

12.6 What the evidence establishes

The benchmark family was designed to isolate repeated cover work. It does not sample arbitrary knot exteriors, establish favorable average behavior, or show that the worst-case exponential constants are small. The explicit index can be a poor practical choice when few covers are useful or memory is scarce. These limitations motivate the broader census and index work in [section 14](#); they do not weaken the complete bounded decision theorems, whose correctness and asymptotic scope come from the proofs.

13 The remaining route to complete recognition

The proved algorithm completely decides a bounded simplification question. To use it as a complete unknot recognizer, one must connect that question to the topology of every relevant

Claim	Basis and present scope
Connected footprint $\leq 3U + 3$	General proof in the stated move category; sharp at $U = 1$ by a separate proof and finite inventory.
$2^{O(U)} \text{poly}(t + B)$ descent	Complete deterministic algorithm with explicit region and trace counting; implemented via shared sleep search.
$2^{O(U+k)} \text{poly}(t + B)$ multiunit descent	General proof using a cited perfect hash family; supplied-coloring DP and positive composition implemented.
Faster covered search	Exact endpoint agreement and paired finite timings on constructed valid solid tori.
General quasi-polynomial ProveIt recognition	Conditional on the additional structural and terminal hypotheses in theorem 13.1 .

Table 4: Proof, implementation, and empirical claims are stated separately.

exterior. This section states a sufficient condition precisely, so that the missing research target is not hidden inside the phrase “small barrier.”

13.1 A conditional closure theorem

Fix the maintained diagram-to-exterior construction and its source binding. Suppose an n -crossing diagram produces, in polynomial time with a verified source binding, a marked triangulation with at most $(n + 2)^{O(1)}$ tetrahedra and polynomial marking bit length. Let $\mathcal{C}$ be a class containing these exteriors and closed under the permitted simplifications. A *sound terminal predicate* P is a computable predicate on members of $\mathcal{C}$ such that $P(T)$ implies that T is a solid torus. Its acceptance must be bound to the actual exterior; a recognizer for an unrelated supplied triangulation is insufficient.

Theorem 13.1 (A sufficient global structural target). *Assume the following two properties for the chosen exterior class and move model.*

1. *The predicate P is decidable in $2^{O((\log(t+B+2))^2)}$ time and is sound for solid tori.*
2. *There is an absolute constant C such that every solid torus $T \in \mathcal{C}$ with $P(T)$ false admits a strict descent with at most $C(\log(t + 2))^2$ total upward moves.*

Then the bounded-descent algorithm and P yield a deterministic $n^{O(\log(n+2))}$ unknot-recognition algorithm for those input diagrams.

Proof. Let t_0 be the initial source size and choose the fixed integer budget $U = \lceil C(\log(t_0 + 2))^2 \rceil$. At each current triangulation first evaluate P . If it accepts, accept the input. Otherwise run complete bounded- U descent. Adopt any independently verified positive endpoint and repeat. If the bounded search exhausts, reject.

Every successful iteration strictly decreases the tetrahedron count, so there are fewer than t_0 such iterations. All replacements preserve the underlying manifold. If the input is knotted, its exterior is not a solid torus, so soundness prevents acceptance by P . If the input is an unknot, every current source is a solid torus in $\mathcal{C}$. Unless P accepts, the second hypothesis supplies a descent within the chosen budget, since the current size is at most t_0 . Complete exhaustion cannot then cause rejection. Finite termination therefore accepts exactly the unknot inputs.

By [corollary 8.7](#), the cumulative marking bit bound increases by at most $t_0 U + O(1)$, remaining polynomial in n . Each descent call costs $2^{O(U)} \text{poly}(t_0 + B) = n^{O(\log(n+2))}$, and the terminal calls have the same form. Multiplying by a polynomial number of iterations

preserves that bound. The maintained exterior binding transfers the conclusion to the original diagram. $\square$

This theorem is a conditional reduction, not a proof of its second hypothesis. It is useful because it permits *any* successful bounded descent at each iteration: the structural hypothesis applies to every resulting solid torus in the closed class. A claim concerning only the existence of one favorable long path would not justify the same greedy choice of intermediate endpoints.

The predicate need not recognize all solid tori. It must cover the solid-torus states that cannot be further simplified within the proposed budget. A terminal family may consequently be much larger than a handful of smallest triangulations. In this move model both the vertices and the boundary triangulation are preserved. These invariants constrain the reachable terminal states and must be part of any valid global theorem.

13.2 Three distinctions that cannot be suppressed

Total upward cost and maximum height. A trace can alternate many upward and downward moves while remaining close to its initial tetrahedron count. A bound on the largest intermediate triangulation therefore does not bound the total number of upward events. The localization and counting theorems charge total upward cost. Converting a height-barrier theorem into this stronger resource bound requires an additional compression argument, potentially using causal components, redundant loops, or a monotone potential. It is not obtained by changing the notation for the parameter.

Descent and disc discovery. The component argument uses additive tetrahedron loss. A disc found after several independent modifications can depend on the combined endpoint geometry and cochain. Its existence need not be witnessed after projecting just one positive component. Any analogous theorem for normal-disc search needs a separate localization invariant for that topological endpoint property. The implementation's shared-cover engine can still accelerate that search, but its complete bounded region is then a supplied hypothesis.

Bounded irreducibility and knot type. A complete failure to descend within budget is an exact statement about the allowed moves. It identifies neither a canonical triangulation nor a nontrivial knot by itself. The terminal and structural hypotheses in [theorem 13.1](#) are what turn bounded failure into a topological decision. Resource-limited failure is weaker still and remains inconclusive.

14 Further research questions and proposed milestones

The questions below separate structural mathematics from implementation work. Each has a concrete target whose success would change the next algorithmic decision.

14.1 Structural questions with direct asymptotic consequences

1. Find a terminal class admitting a small total-upward theorem. For the exact finite exterior class produced by ProveIt, identify a sound terminal predicate P and prove, or refute by a family, the second hypothesis of [theorem 13.1](#). Boundary triangulation and vertex data must be retained in the statement. A useful intermediate result would establish $U = O(\log^2 t)$

for a natural subclass such as a controlled family of layered or hierarchy-derived solid tori. The milestone is a uniform theorem over that subclass, with its constants and encoding costs explicit.

2. Relate height barriers to total upward cost. Let h bound the excess tetrahedra along a simplification path. What extra condition on that path, its dependency graph, or its hierarchy controls the minimum total upward cost by $\text{poly}(h, \log t)$? A negative family with small h and large minimum total upward cost would also be valuable: it would rule out a tempting but invalid route to the required bound. Recording both parameters in future searches is a necessary first experiment.

3. Determine the optimal localization function. Define $f(U)$ as the least universal radius guaranteeing a connected first-descent footprint whenever descent with at most U upward events exists in the stated category. This report proves $f(U) \leq 3U + 3$ and $f(1) = 6$. Is the linear coefficient three asymptotically sharp, or do geometric constraints force extra birth arcs and improve the incidence estimate? Any proposed lower-bound family must rule out *all* alternative low-cost simplifications, not only exhibit a large chosen trace. The six-cell proof illustrates the required strength.

4. Generalize the additive objective carefully. Which other integer potentials admit a componentwise additive loss and a bounded-per-event incidence census? A potential involving tetrahedra, specific normal coordinates, or a hierarchy measure could make more terminal states accessible without leaving the causal framework. The target is an abstract theorem with explicit local input/output sizes, backward persistence, and a positive-component argument. Nonadditive normal-disc existence should first be tested for counterexamples to naive projection.

5. Extend the move set without losing the proof contract. Boundary moves, vertex-changing moves, or 2–0 reductions may overcome the preserved invariants of the present model. Each addition changes the legality and input/output census. One must establish whether every enabling dependency still consumes a born cell, or whether a richer graph is needed. The strict-simplicial suspension example gives a concrete diagnostic: global missing-simplex tests create dependencies invisible to tetrahedron births. A useful result would quantify localization for a larger, precisely specified move system rather than appeal to Pachner connectivity without resource bounds.

14.2 Algorithmic questions suggested by the new theorems

6. Complete the deterministic multiunit implementation. Implement a reproducible perfect-hash-family generator, stream authenticated positive items through the supplied-coloring DP, reconstruct the selected certificates, and call the disjoint-trace composer. An end-to-end negative answer must certify that every required coloring and item family was exhausted. The first milestone is agreement with unrestricted small-instance search under a *joint* upward budget. Enumerating components separately and greedily repeating descents is not a substitute for this comparison.

7. Reduce index memory while preserving exact cover semantics. The full inverted index removes repeated search but stores all connected covers. Can a compact decision diagram,

canonical-parent search oracle, or incremental connected-completion structure represent $\{R : F \subseteq R\}$ exactly with substantially lower practical memory? An admissibility oracle must distinguish actual footprints from their possible connecting covers. It must also support resource-capped failure without misreporting complete exhaustion. Compare streaming restart, explicit index, and compact index on both positive and fully exhausted instances, including setup time and peak memory.

8. Prove a stronger safe quotient of continuation states. The present deduplication saves endpoint queries but retains continuation frames. A geometric isomorphism alone forgets original-source supports and sleep history. Determine the smallest additional data making two frames future-equivalent under the remaining upward budget and cover family. A proof of a right congruence would justify further pruning. Until then, endpoint equality remains evidence about outputs, not permission to merge all histories.

9. Improve local geometry updates and marking storage. Global validation and rebuilding are polynomial but dominate small traces. Maintain affected edge stars, dual incidences, and cochain transport using stable local frames and persistent data structures. The acceptance contract must remain independent: a faster producer should still emit certificates checked without calling its own move or transport routines. Test every accepted local frame, particularly downward belt relabelings, as part of the update invariant rather than relying only on the producer’s canonical examples.

10. Tighten the trace-count constants and use budget structure. The 11-symbol encoding deliberately permits many impossible commitments. Can local compatibility, first-descent balance, or birth-graph sparsity give a smaller exponential base without costly per-state reasoning? A second parameter such as pathwidth of the consumed region may support a different dynamic program. The milestone is a proved count and a measured crossover, not a branching-factor estimate based only on successful fixtures.

14.3 Evidence, verification, and integration questions

11. Build a heterogeneous barrier census. Extend the current constructed solid-torus controls with genuine diagram exteriors, independently recognized hard solid tori, local-minimum triangulations, and adversarial cases from published Pachner-graph exploration. Record total upward cost, maximum height, minimum actual footprint, component structure, index size, peak memory, and complete or capped status. Retain unsuccessful cases and certificate replay. Such a census can guide conjectures about [theorem 13.1](#); it cannot prove the required asymptotic bound on its own.

12. Certify negative bounded search and formalize the core. Positive traces already have an independent geometric checker. Can complete bounded failure be exported as a compact exploration DAG whose checker verifies initial state, all outgoing alternatives, resource budgets, commutation justifications, and terminal rejection? Separately, formalize the small core consisting of backward persistence, component projection, footprint connectivity, and the incidence theorem. A formalization should start from explicit generalized face pairings and local permutations; ordinary strict simplicial complexes have different legality rules. This would give the repository a precise trust boundary for future optimizations.

14.4 A practical order of work

The immediate integration milestone is the additive cover search with independent replay and the default radius $3U + 3$. The next implementation milestone is complete multiunit color coding and a broader measured census. In parallel, the highest-value mathematical task is the terminal-class and total-upward structural theorem. Better constants and data structures make the current bounded primitive more useful, while that structural theorem is what could close the project's specific quasi-polynomial recognition argument.

15 Deliverables and integration contract

The archive contains the complete article sources and compiled PDF, the new producer and verifier modules, causal certificate utilities, packing prototype, fixtures, targeted tests, exact raw benchmark records, positive certificates, independent audit results, and a self-contained reproduction snapshot. The snapshot combines the pinned native tree with only the previously absent incoming dependencies and the additive files from this report. It is not an instruction to overwrite the current repository with the whole snapshot.

The integration script defaults to a dry run. It checks recorded native prerequisite hashes and refuses conflicting destination files; applying the overlay only creates absent files or accepts byte-identical existing ones. A manifest identifies native baseline files, incoming dependencies, and this report's additions separately. Reproduction scripts retain new outputs separately from the recorded evidence. Checksums make the delivered bytes auditable, while mathematical correctness rests on the proofs and geometric verification rather than on those checksums.

The resulting contribution is a complete fixed-parameter simplification theorem, its multiunit strengthening, and a measured implementation of shared local search with independently checked witnesses. The project-wide recognition consequence is stated with its remaining assumptions in [theorem 13.1](#), so it can be pursued as a precise research problem.

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